

BHAVATARINI THANGARAJU

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EDUCATION

Illinois Institute of Technology, Chicago, Illinois

January 2024 - December 2025 (Expected)

Master of Data Science

GPA:3.9

Coursework: Data Structures and Algorithms, Operating Systems, Cloud Services, Machine Learning (ML), Big Data Technologies

WORK EXPERIENCE

Software Engineering Intern - Machine Learning

January 2025 - May 2025

Labelmaster, Chicago, Illinois, USA

- Conducted exploratory data analysis (EDA) on CRM data using Python to uncover patterns that improved feature selection and data quality
- Orchestrated data cleaning and NLP with NLTK and SentenceTransformer to extract job title semantics, increasing model accuracy to 90%
- Built scalable ML pipelines with XGBoost to impute missing profiling fields across 350K+ records, raising data completeness to 95%
- Applied KMeans and DBSCAN clustering to segment accounts for personalized outreach, boosting marketing engagement efficiency by 50%

Software Engineer

July 2022 - December 2023

Bosch Global Software Technologies, Coimbatore, Tamil Nadu, India

- Owned end-to-end development of a core Spring Boot microservice with backend logic, data modeling, and Angular-based frontend
- Implemented JWT authentication and role-based access control using Spring Security and API Gateway, securing 30+ endpoints and reducing unauthorized access incidents by ~40%
- Designed and developed scalable RESTful APIs using Spring Boot, enabling seamless integration for 5+ backend services and 20 client modules
- Designed and maintained high-throughput ETL pipelines using Java and Oracle PL/SQL to clean, transform, and load 60M+ legacy records, optimizing schema design for faster query execution and efficient data retrieval

Associate Software Engineer

July 2021 - June 2022

Bosch Global Software Technologies, Coimbatore, Tamil Nadu, India

- Configured Apache Tomcat server with load balancing and failover support, ensuring 99.9% uptime and reduced latency during peak traffic
- Automated daily data syncs using Oracle packages to synchronize external systems with internal databases, improving data reliability
- Optimized high-volume search endpoints by rewriting SQL queries with indexing and Java-based pagination, reducing response time by 98.3%
- Mentored junior developers through code reviews, documentation, knowledge transfers strengthening team collaboration and code delivery

Software Engineer Intern

September 2020 - June 2021

Bosch Global Software Technologies, Coimbatore, Tamil Nadu, India

- Developed a Python based rule-based chatbot microservice to handle user FAQs, reducing manual support load by approximately 40%
- Boosted unit and integration test coverage by 80% using JUnit, JSuite, and Mockito, increasing code reliability and test automation confidence
- Implemented and maintained CI/CD pipelines using Jenkins for frontend and backend test environments, reducing deployment time by 60%

PROJECTS

ClinicPro - Patient Management System (Java, Spring Boot, REST, gRPC, Kafka, PostgreSQL, Docker, JWT, AWS CDK)

- Architected a modular Spring Boot backend with five microservices (auth, patient, billing, analytics, gateway), integrating JWT-secured REST/gRPC APIs, Kafka-based event streaming, and centralized routing via Spring Cloud Gateway
- Dockerized all services and provisioned production-grade cloud infrastructure using AWS CDK (Java), deploying to LocalStack (ECS Fargate, RDS, MSK, ALB) with seamless portability to AWS via CloudFormation templates

Gesture-Driven Assistive Communication System (Python, OpenCV, PyTorch, Flask, CNN, pytsx3)

- Created a real-time eye gesture recognition system using Python, OpenCV, PyTorch, and Flask; generated and augmented a custom dataset (10,000+ images), achieving 95% CNN classification accuracy after image preprocessing and feature extraction
- Translated classified gestures into Morse code from video input based on gesture duration and converted the output to audio using pytsx3, enabling non-verbal communication for users with mobility impairments

Dynamic Pricing for Airbnb (Python, Scikit-learn, FastAPI, Streamlit, K-Means, Random Forest, Logistic Regression)

- Built a machine learning-driven pricing model using Python, Scikit-learn, Streamlit, and FastAPI; applied data cleaning, outlier removal, Random Forest feature selection, and K-Means clustering on geolocation, improving model precision by 25%
- Trained and deployed a Logistic Regression model, achieving 90% prediction accuracy for Airbnb rental prices

SKILLS

Programming Languages & Frameworks: Java, Python, R, Angular, NumPy, Pandas, TensorFlow, Matplotlib

Web & Data Technologies: Oracle, PostgreSQL, MongoDB, Hive, Spark, Hadoop, Snowflake, HTML, CSS, Typescript

Cloud & DevOps Tools: AWS (EMR, S3, Lambda, Redshift, Glue, Kinesis, DynamoDB, SNS, SQS, ECS, RDS), Kubernetes, Apache Airflow, Git

CERTIFICATIONS

- AWS Certified Cloud Practitioner - Amazon Web Services (2024)